# Shubham Gupta

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## SUMMARY

ML Developer specialising in VFX production pipelines, with two and a half years at Rising Sun Pictures delivering production-ready ML solutions for complex media application problems across multiple feature film productions. Focused on facial performance transfer, digital head replacement, and ML-driven shot solutions, working across the full pipeline from model development through Flow (formerly ShotGrid)-tracked delivery. Credited on the Academy Award-nominated VFX pipeline for Sinners (2026).

## EXPERIENCE

**Rising Sun Pictures** · Machine Learning Developer · August 2024 – Present

- **Academy Award–Nominated Production:** Contributed to the VFX pipeline for *Sinners*, nominated for Best Visual Effects at the 2026 Academy Awards. Worked on digital face replacement combining traditional and ML-driven techniques, and co-developed external-facing presentation materials on the ML pipeline used in production.
- **Performance Transfer:** Contributed to the pipeline enabling a hero actor's facial performance to be applied to a stunt double's body across long takes (~500 frames each), helping build the workflow at RSP.
- **ComfyUI Pipeline Development:** Well-versed in managing ComfyUI-based generative pipelines, having purpose-built numerous custom ComfyUI nodes to support production workflows.
- **Performance Capture Workflow:** Developed the end-to-end pipeline connecting ML model outputs to Maya facial rig parameters, enabling predicted expression values to drive animation curves on a timeline, and extended it with a gaze estimation module to resolve missing eye-gaze results using fine-tuned and pre-trained models.
- **Shot Stitching / Outpaint / SloMo Solutions Using ML:** Delivered custom, purpose-built ML solutions for shot stitching, outpainting, and slow-motion generation across several projects, including blending two separate takes on a single shot. These are traditionally expensive, compositing-heavy problems, but ML approaches worked well even with fewer source frames and shorter timeframes, delivering at a fraction of the typical cost, working closely with department supervisors from requirements through to final delivery.
- **Industry Presentations:** Contributed to external-facing presentations on RSP's ML capabilities, preparing materials alongside department supervisors for industry audiences, and presented the ML team's work to students during Day in the Life Of (DITLO) sessions.
- **Data at Scale:** Comfortable managing and processing large-scale datasets (up to 500,000 images per project) across multiple concurrent productions, with strong working knowledge of the systems and infrastructure needed to handle data at scale.

**InsightFactory.ai** · Data Scientist · July 2021 – July 2024

- Reduced operational costs by 50% for a major mining client by architecting a fleet-wide engine failure prediction system with closed-loop monitoring and iterative model refinement.
- Cut rail disruptions by 20% through a predictive maintenance model with ongoing accuracy tracking against real-world outcomes.
- Built high-throughput SQL pipelines on Databricks processing millions of rows of time-series data daily, supporting scheduled production model execution.
- Delivered executive-facing Power BI analytics platform integrating multi-source data; directly informed resource allocation decisions at leadership level.

**The University of Adelaide** · Teaching Assistant · Research Assistant · Aug 2020 – Dec 2021

- Led Computer Vision workshops for 50+ students; Co-authored a peer-reviewed NLP publication (PACLIC 2022, arXiv:2109.11199) on abstractive multi-document summarisation.

**Australian Institute of Machine Learning** · Research Project · July 2020 – June 2021

- Developed and evaluated generative models (GANs, CGANs, StackGANs) for foreground image synthesis; collaborated with Dr. Rafael Felix Alves.

## EDUCATION

**Master of Data Science** · The University of Adelaide · GPA 6.467 / 7 · 2019 – 2021

## SKILLS

**VFX & Pipeline:** Flow (formerly ShotGrid), Maya (rig integration, animation export), Nuke, ComfyUI, on-premise farm pipelines

**ML & Deep Learning:** PyTorch, GANs, VAEs, Neural Rendering, Gaussian Avatars, performance transfer, gaze estimation, fine-tuning

**Programming:** Python, PySpark, SQL, C++, MATLAB

**MLOps & Infra:** Docker, Git, Azure, Databricks, on-premise server pipelines, DevOps, Agile

**Data & Viz:** Pandas, NumPy, Power BI, DAX

**Languages:** English (Fluent), Hindi (Fluent)

## AWARDS & ACTIVITIES

- Contributed to the VFX pipeline for Sinners, nominated for Best Visual Effects at the 2026 Academy Awards
- Adelaide Graduate Award (May 2021) · University of Adelaide Research Scholarship (Dec 2020)
- Volunteer Kitchen Assistant: Meals on Wheels, Adelaide (June 2022 – 2025)